

PROJECT S.A.M

Secure Digital Asset Management

Protecting Digital Assets with Blockchain-Inspired Auditing
and AES Encryption.

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| THE SECURITY GAP



The Problem

Traditional file storage systems lack native end-to-end encryption, leaving data vulnerable.



The Risk

Server-side breaches and lack of tamper-proof audit trails expose critical file interactions.



The Need

A unified vault combining high-security encryption with proactive monitoring.

| METHODOLOGY & APPROACH

-  **Architecture:** Spring Boot micro-monolith focusing on a "Security First" design pattern.
-  **Encryption:** Robust AES-128 bit encryption enforced for every byte of uploaded data.
-  **Integrity:** Blockchain-inspired logging; security events are SHA-256 hashed and chained.
-  **Frontend:** Glassmorphic UI with Three.js WebGL backgrounds for immersive visuals.
-  **Database:** H2 embedded database utilized with encrypted storage pathing.

| RESULTS & OUTCOMES

- 🛡️ **Successful Deployment:** A fully functional web-based vault for depositing and withdrawing assets with total privacy.
 - 🛡️ **Zero-Plaintext Storage:** No sensitive files are stored in their original form; everything is encrypted at the byte level.
 - 🛡️ **Secure Auditing:** Verified tamper-proof logs that can detect if an audit entry has been altered, ensuring accountability.
 - 🛡️ **User Engagement:** Premium UI/UX that makes high-level security management accessible and visually engaging.
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CONCLUSION

Project Impact

S.A.M successfully demonstrates that heavy cybersecurity measures can be integrated seamlessly with modern, premium web experiences without compromising performance.

Key Achievements

We met all core objectives: robust byte-level encryption, role-based access control, and forensic blockchain auditing are live on the platform.

Future Roadmap

Expansion to multi-cloud storage clusters and integration of hardware-based MFA (U2F/FIDO2) for physical key authentication.